A Microfluidic Immunomagnetic Platform for NK Cell Isolation from Whole Blood

[1]Haehee Han, [2]Yongje Jeon, [2]Gyuri Park, [3]Seungyoun Chae, [1]Yeonwoo Kang, [2]Jaewon Park*, [1]Sungkyu Seo*
[1]Department of Electronics and Information Engineering, Korea University, Sejong, Korea, [2]Department of Electro-Mechanical Systems Engineering, Korea University, Sejong, Korea, [3]Department of Biotechnology and Bioinformatics, Korea University, Sejong, Korea
E-mail: netjay@korea.ac.kr, sseo@korea.ac.kr

전혈 기반 NK 세포 분리를 위한 미세유체 면역자기 플랫폼

한해희 [1], 전용제 [2], 박규리 [2], 채승연 [3], 강연우 [1], 박재원 [2,*], [1] 서성규 [1,*]
[1] 고려대학교 전자·정보공학과, [2] 고려대학교 전자·기계융합공학과, [3] 고려대학교 생명정보공학과
E-mails: netjay@korea.ac.kr (박재원), sseo@korea.ac.kr (서성규)

Abstract
Isolation of natural killer (NK) cells from whole blood is essential for immunological analysis, yet conventional methods typically require large sample volumes and rely on expensive, specialized equipment coupled with labor-intensive preprocessing steps. Here, we present a microfluidic NK cell separation chip based on negative immunomagnetic selection that enables processing of a small volume (20 µL) of human whole blood. Separation performance was systematically evaluated by flow cytometry as a function of flow rate and magnetic field strength. Optimal performance was achieved at 10 µL/min with a 2330 G magnetic field, yielding an NK cell purity of up to 77%. This work demonstrates the feasibility of NK cell isolation from small volumes of whole blood using a microfluidic platform, highlighting its potential for point-of-care immunological applications.

Keywords: ***Microfluidics; NK cells; Immunomagnetic separation; Whole blood; POC (Point-of-care)***

1. Introduction

Natural killer (NK) cells are key components of the innate immune system and serve as important indicators of immune status. Reliable isolation of NK cells from blood is therefore essential for immunological analysis and diagnostics.

Conventional NK cell isolation methods, such as fluorescence-activated cell sorting (FACS) and magnetic-activated cell sorting (MACS), provide high performance but require bulky equipment, large sample volumes, and skilled operators, limiting their use to centralized laboratories.

In this study, we present a microfluidic chip that isolates NK cells from only 20 µL of whole blood using a syringe pump and permanent magnets. The system demonstrates stable and reproducible separation with minimal user intervention, offering a simple and practical NK cell isolation platform for point-of-care immunological diagnostics.

2. Materials and Methods

2.1 Chip Design

The microfluidic chip operates based on a negative immunomagnetic separation principle, in which non-target cells in whole blood are selectively labeled with magnetic beads and captured under an external magnetic field, while unlabeled NK cells are transported toward the outlet.

The chip has an overall dimension of 26 × 38 mm with a total internal volume of approximately 65 µL. The main channel was designed as a serpentine flow path with a width of 400 µm and a height of 180 µm to ensure stable flow of whole blood samples under high cellular load conditions. Magnetic trapping regions were systematically incorporated at each turning point of the serpentine channel with a uniform width of 1.5 mm, resulting in a total trapping volume of 15.2 µL. To support this separation mechanism, a multi-magnet configuration was used to generate a stable and uniform magnetic field gradient across all trapping regions, thereby maximizing trapping efficiency.

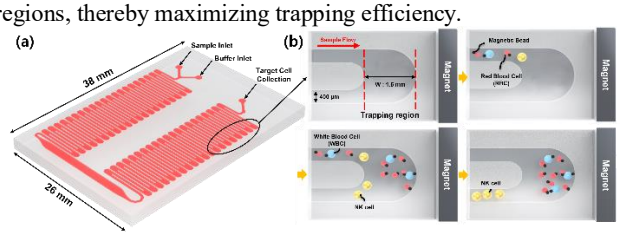

Figure 1. (a) Design of a microchip for NK cell separation, (b) Mechanism of NK cell separation

2.2 Sample Processing and Separation Procedure

Whole blood samples were obtained from healthy adult donors and processed using a commercial NK cell isolation kit following the manufacturer's recommended antibody and magnetic bead labeling protocol. The labeled whole blood samples were then introduced into the microfluidic chip using a syringe pump at three different flow rates (5, 10, and 15 µL/min).

For each experiment, 20 µL of whole blood sample was first injected to allow magnetically labeled non-target cells to be captured within the magnetic trapping regions, followed by 65 µL of buffer to hydrodynamically drive unlabeled NK cells toward the outlet.

2.3 Flow Cytometry Analysis

Separated NK cells were analyzed by flow cytometry (FACS) using fluorescently labeled CD56 and CD3 antibodies. NK cell

purity was defined as the proportion of $CD56^{+}CD3^{-}$ cells among the total cells collected at the outlet. The number of recovered NK cells was calculated based on the total cell count and the corresponding purity value.

3. Results and Discussion

The proposed microfluidic NK cell separation chip enabled effective NK cell isolation using a small volume of whole blood (20 µL). The entire separation process was completed within approximately 10 minutes, representing a substantial reduction in both sample volume and processing time compared with conventional methods. These results demonstrate the feasibility of rapid immunological analysis using small blood volumes and highlight the potential of the platform for point-of-care applications.

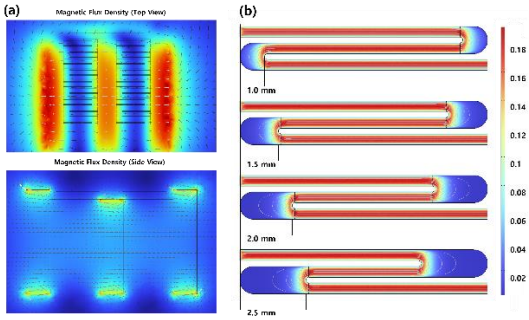

Figure 2 Magnetic flux density simulations applied to the chip: (a) top view and side view; (b) flow simulations with varying trapping region widths

Under whole-blood operating conditions, separation performance was governed by the interplay between the channel geometry, hydrodynamic transport, and magnetic retention force. In particular, the geometry of the trapping regions played a critical role in ensuring stable separation performance and sufficient blood processing capacity. Flow simulations revealed that when low-velocity regions occupied more than 50% of the trapping region area, frequent air bubble entrapment occurred during whole-blood flow through the channel, thereby reducing effective trap utilization. Conversely, an insufficient trapping width constrained the effective magnetic capture area and limited the amount of whole blood that could be processed in a single run. Based on combined simulation and experimental results, a trapping region width of 1.5 mm was identified as the optimal condition, enabling stable flow, efficient capture of magnetically labeled non-target cells, and smooth recovery of unlabeled NK cells.

With the trapping region geometry fixed at 1.5 mm, separation performance was further evaluated as a function of flow rate and magnetic field strength, which governs residence time and magnetic retention in immunomagnetic separation systems. Among the tested flow rates, the most stable separation performance was observed at 10 µL/min, yielding an average NK cell purity of 59.8% with a low coefficient of variation (CV = 14.9%), indicating high reproducibility across repeated experiments. The low CV suggests reduced sensitivity to experimental handling, indicating that reliable NK cell separation can be achieved with minimal dependence on operator expertise. In contrast, at 5 µL/min, the average purity decreased to 28.6% with a markedly higher CV (84.6%), reflecting poor reproducibility. At 15 µL/min, the average number of recovered NK cells increased to 2,657 cells, but the average purity decreased to 49.8%, indicating reduced separation accuracy due to partial displacement of magnetically labeled cells by increased hydrodynamic forces. These results demonstrate that separation performance is strongly governed by a balance between hydrodynamic transport and magnetic retention forces.

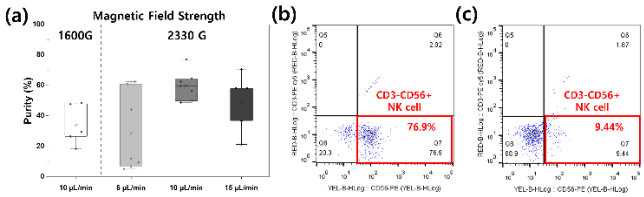

Figure 3 FACS analysis of NK cell separation : (a) Separation purity depending on magnetic field strength and flow rate; (b,c) Representative FACS plots at flow rates of 10 and 5 µL/min.

The effect of magnetic field strength was then evaluated at a fixed flow rate of 10 µL/min. Using a 1600 G magnet resulted in an average NK cell purity of 34.1%, whereas increasing the magnetic field strength to 2330 G improved the average purity to 59.8%, confirming that sufficient magnetic force is essential for stable retention of magnetically labeled non-target cells. These findings demonstrate that the multi-magnet configuration provides a robust magnetic environment capable of sustaining stable separation under whole-blood conditions.

These results delineate a practical operating window for the proposed chip, in which stable and reproducible NK cell separation is achieved under whole-blood conditions through experimentally validated balance between channel geometry, hydrodynamic transport, and magnetic retention.

4. Conclusion

In this study, we developed a microfluidic NK cell separation chip based on negative immunomagnetic selection that enables reliable NK cell isolation from a small volume of whole blood (20 µL). By integrating optimized magnetic trapping geometry with simple operating conditions, stable separation performance was achieved without reliance on bulky instrumentation or complex workflows. Quantitative evaluation using flow cytometry demonstrated that NK cell purity is strongly governed by flow rate and magnetic field strength, with optimal and highly reproducible performance obtained at 10 µL/min and 2330 G. Importantly, the low variability observed under optimal conditions indicates reduced dependence on operator expertise, supporting practical use by non-expert users. These results suggest that the proposed platform can serve as a compact and robust NK cell separation module for point-of-care immunological diagnostics and decentralized analytical systems.

Acknowledgments

This study was supported by the National Research Foundation of Korea (NRF; Grant Nos. RS-2024-00353675, RS-2025-00562827), the IITP ITRC program and COMPA funded by MSIT (Grant Nos. IITP-2025-RS-2023-00258971, 2710084652), the Korea Institute of Marine Science and Technology Promotion (KIMST) funded by MOF (Grant No. 20210660), and the Ministry of SMEs and Startups (MSS; Grant No. RS-2025-02310307).